Session 5: Solutions and facilitation notes

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Exercise numbers follow `Practicals/s5_exercises.md`. All FLUID-ICU numbers come from `Course/Data/key_results.md` (simulated data).

Exercise 1: Classify the variables

#	Variable	Answer	Explanation
a	Age (years)	**Continuous**	Measured on a scale. Keep the exact number on the CRF (or collect date of birth).
b	Sex	**Binary**	Two categories. Coding 1/2 does not make it numerical.
c	Glasgow Coma Scale	**Ordinal**	Ordered scores (3–15) with unequal steps. Often *analysed* like a number, which is acceptable if stated, but it is ordinal by nature.
d	Source of sepsis	**Nominal**	Several categories, no natural order.
e	Number of vasopressors	**Count**	Whole numbers 0–3. Accept "ordinal" if the participant argues it well; not continuous.
f	Cumulative fluid balance (ml/kg)	**Continuous**	Can be negative or positive, with any value on the scale. The Paper 1 primary outcome.
g	Alive at day 28	**Binary**	Yes/no.
h	Time to death, censored at day 28	**Time-to-event**	Two pieces of information: how long, and whether the event happened. Analysed with Kaplan–Meier / Cox.
i	Pain none–severe	**Ordinal**	Ordered categories.
j	Blood group	**Nominal**	No order.

Bonus: any continuous variable (age, fluid balance, lactate). You can always group later, but you cannot recover the exact value from a group.

Common errors: GCS called continuous; time-to-death called continuous (ignores censoring); number of vasopressors called continuous.

Exercise 2: Fix the bad CRF items (model answers)

#	Bad item	Problem	Model rewrite
a	Fluid given: ____	No unit, no time window, no definition (crystalloid? blood? drug diluents?)	"Total IV fluid received from ICU admission to 24 h (ml), including crystalloids, colloids and drug diluents: [ ][ ][ ][ ][ ] ml □ not recorded"
b	Weight: ____	No unit; measured or estimated? when?	"Body weight at ICU admission (kg): [ ][ ][ ].[ ] □ measured □ estimated □ not available"
c	Comorbidities: _____	Free text: impossible to analyse, spelling varies	Tick list: "Comorbidities present at admission (tick ALL that apply): □ diabetes □ chronic kidney disease □ heart failure □ COPD □ cirrhosis □ cancer □ none □ unknown"
d	Lactate high? Y / N	Opinion ("high" by whose cut-off?), loses the value, no time point	"Serum lactate at ICU admission (mmol/L): [ ][ ].[ ] □ not done" (define "high" later, in the analysis)

Accept any version that passes the four checks (unit, time point, one reading, unknown option). Praise answers that notice that collecting the *number* is better than collecting a yes/no.

Exercise 3: Choose the right summary

#	Variable	Shape / type	Summary	Graph
a	Age, bell-shaped	Symmetric continuous	**Mean (SD)**	Histogram
b	ICU length of stay	Right-skewed	**Median (IQR)**	Histogram or boxplot
c	Died in hospital	Binary	**n (%)**	Bar chart
d	CRP	Right-skewed	**Median (IQR)**	Histogram / boxplot
e	Source of sepsis	Nominal	**n (%) per category**	Bar chart (not 3-D pie)
f	SBP by treatment group	Symmetric, two groups	**Mean (SD) per group**	Side-by-side boxplots

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Exercise 4: Judge a bad figure

1. Each bar is the **mean** fluid balance per group. The error bar is the **standard error (SE)**, shown in one direction only. Missing: the individual patients, the spread (SD or IQR), the shape (skew), the n per group on the figure, and what the error bar represents.

2. Any two of: the data are **skewed**; **one extreme patient** (160 ml/kg) pulls the usual-care mean up; the groups **overlap** a lot; the median; the number of patients; negative values.

3. Dots (one per patient) plus a boxplot per group (see the "dots + box" slide), y-axis labelled with units, n under each group. You gain: shape, spread, outliers and overlap become visible, and the reader can judge the comparison.

4. The truncated axis makes a 2-percentage-point difference (31% vs 29%) look like a three-fold difference. Fix: start bars at 0 and label the values (or show the difference with its 95% CI).

5. (i) The author chose the two scales, so the lines can be made to cross or move together wherever wanted; (ii) two trends over time moving together is **correlation, not causation** (other changes may have happened at the same time). Better: two panels sharing the time axis, and an appropriate design (e.g. interrupted time series, Session 3).

Exercise 5: [FLUID-ICU] Spot the problems in the raw FLUID-ICU data

(See `Course/Data/cleaning_log.md` for the full log.)

#	What is wrong	What the script does
a	One variable coded many ways (and a leading space)	Recode to `Female` / `Male`
b	Dates in different formats (03/04: March or April?)	Read each format explicitly and convert to one date format (YYYY-MM-DD)
c	ICU C reported lactate in mg/dL, not mmol/L (about 9 times larger; found by comparing medians by ICU)	Divide ICU C values by 9.01 (1 mmol/L = 9.01 mg/dL); confirm with the site laboratory
d	`999` is a missing-value code, not a lactate	Set to missing (NA)
e	Impossible age (outside the allowed range)	Set to missing, send a query to the site; keep the patient
f	Duplicate record	Check the two rows are identical, keep one copy (623 rows become 620 patients)

Bonus: editing by hand is invisible and not reproducible. A script plus a log lets anyone check and repeat every change. The log records the problem, how many records, what was done, and why (plus queries sent).

Exercise 6: [FLUID-ICU] Read the FLUID-ICU Table 1 (whole class)

All numbers from `Course/Data/key_results.md` (simulated data).

1. Negative balance n = 245, positive n = 373, overall 618. Two of the 620 enrolled patients have no computable day-3 fluid balance (an impossible weight or fluid output was set to missing during cleaning), so they cannot be placed in a group.

2. Expected answers: mean (SD) for age, which is roughly symmetric (64.6 vs 61.4 years); median (IQR) for SOFA, a skewed score (6 (4 to 8) vs 8 (6 to 10)); median (IQR) for lactate, which is right-skewed (2.2 (1.4 to 3.0) vs 2.9 (1.9 to 4.0) mmol/L); n (%) for vasopressor, ventilation, heart failure and sex.

3. Severity: SOFA 8 vs 6; vasopressor 205/373 (55.0%) vs 61/245 (24.9%); mechanical ventilation 216/373 (57.9%) vs 97/245 (39.6%). (Heart failure goes the other way: 43/373 (11.5%) vs 54/245 (22.0%), because clinicians give less fluid in heart failure.)

4. Lactate: 91 missing among the 618 (93 of all 620 enrolled, 15.0%).

5. Model sentences: "Of 620 enrolled patients, 618 had a computable day-3 fluid balance, of whom 373 (60.4%) had a positive balance. Patients with a positive balance were more severely ill at admission (median SOFA 8 vs 6; vasopressor 55.0% vs 24.9%; mechanical ventilation 57.9% vs 39.6%)."

6. Larger: the positive-balance group was sicker, and sicker patients die more, so the crude association mixes fluid with severity (confounding by indication). Confirmed in Session 6: crude OR 2.49 vs adjusted 1.62.

Exercise 7: [FLUID-ICU] Your manuscript question as a PECO sentence

In class, Q1 is done together; each participant later completes the table for the question they chose.

Q	Outcome (type)	Exposure (type)	Planned comparison / model
Q1	Death by day 28 (binary; also time-to-event)	Positive balance by day 3 (binary)	Chi-square; logistic regression; Kaplan–Meier + Cox
Q2	AKI, KDIGO ≥ 1 (binary)	Cumulative balance, ml/kg (continuous, per 10 ml/kg)	Chi-square by group; logistic regression
Q3	Ventilator-free days to day 28 (count 0–28, skewed, piled at 0 and 28)	Cumulative balance (continuous)	Mann–Whitney; linear regression
Q4	Death by day 28 (binary)	Lactate at admission, mmol/L (continuous, 15% missing)	Logistic regression; complete case vs multiple imputation
Q5	ICU length of stay in survivors (continuous, right-skewed)	Positive balance (binary)	Mann–Whitney; linear regression on log scale

Example (Q1): "In adults with sepsis admitted to the ICU, is a positive cumulative fluid balance by day 3, compared with a negative balance, associated with higher 28-day mortality?"

Exercise 8: Proposal builder 5 (what a good answer looks like)

Variables table

Role	Variable	How measured	Type
Primary outcome	Death by day 28	Hospital records / phone call at day 28	Binary
Secondary outcome	Cumulative fluid balance at day 5	Fluid charts, ml/kg of admission weight	Continuous
Intervention	Restrictive vs usual fluid strategy	Randomised allocation	Binary

Role	Variable	How measured	Type
Baseline / confounders	Age, SOFA score, lactate at admission, source of sepsis	CRF at ICU admission	Continuous, ordinal-like score, continuous, nominal

Sample size: 40% vs 30% mortality, alpha 5% two-sided, power 80% gives 356 per group (spreadsheet sheet 2). Dividing by (1 − 0.10) gives 396 per group, 792 in total.

Analysis plan: describe with mean (SD) / median (IQR) / n (%); compare 28-day mortality with a chi-square test; report the risk ratio and risk difference with 95% CIs; fluid balance compared with Mann–Whitney and reported as median (IQR) per group. (Observational designs: add "adjusted for age, SOFA and lactate using logistic regression".)

Quick check for the instructor (while modelling, and when reading drafts sent by email):

Element	Good enough	Needs work
Variables table	Outcome, exposure and ≥ 3 confounders, all with a type	Confounders that are actually outcomes; no types
Mini-CRF	6–10 items, units, time points, unknown option	> 15 items; free text; no units
Data dictionary	3–5 rows with codes and allowed ranges	Names with spaces, no codes
Sample size	4 inputs with a stated source	"All patients in one year"
Analysis plan	Summary + test + effect measure with 95% CI	"Data will be analysed with SPSS"

Data management (good enough): EDC with range and logic checks at entry (or paper with double entry of key variables); coded study IDs with the linking log kept separately and locked; audit trail/cleaning log; raw export read-only, dated versions, encrypted 3-2-1 backups; data-sharing statement in the consent form.